

The image displays four Feynman diagrams, each representing a different way to cut a box diagram with a vertical dashed red line. The diagrams are summed and equated to a mathematical expression involving a double-cut integral.

The first diagram shows a box diagram with external momenta  $p_1$  (top),  $p_2$  (bottom),  $k$  (left), and  $k-p_1$  (right). The internal momenta are  $k-p_2$  (top-left),  $p_{12}-k$  (bottom-left),  $k$  (top-right), and  $k-p_1$  (bottom-right). The cut is a vertical dashed red line.

The second diagram shows a box diagram with external momenta  $k-p_1$  (top),  $p_2$  (bottom),  $k$  (left), and  $k-p_1$  (right). The internal momenta are  $k-p_2$  (top-left),  $p_{12}-k$  (bottom-left),  $k$  (top-right), and  $k-p_1$  (bottom-right). The cut is a vertical dashed red line.

The third diagram shows a box diagram with external momenta  $p_1$  (top),  $k-p_2$  (bottom),  $k$  (left), and  $k-p_1$  (right). The internal momenta are  $k-p_2$  (top-left),  $p_{12}-k$  (bottom-left),  $k$  (top-right), and  $k-p_1$  (bottom-right). The cut is a vertical dashed red line.

The fourth diagram shows a box diagram with external momenta  $k-p_1$  (top),  $k-p_2$  (bottom),  $k$  (left), and  $k-p_1$  (right). The internal momenta are  $k-p_2$  (top-left),  $p_{12}-k$  (bottom-left),  $k$  (top-right), and  $k-p_1$  (bottom-right). The cut is a vertical dashed red line.

The sum of these four diagrams is equal to the following expression:

$$= \text{disc}_{p_1^2} \text{disc}_{p_2^2} \int \frac{d^d k}{(2\pi)^d} \frac{N_{\boxtimes}^{qq'} \delta_{q^2}^+(p_{12}-k) \delta_0^+(k)}{p_1^2 p_2^2 (k-p_2)^2 (k-p_1)^2}$$